

Multivariable Calculus

Unit 4
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4	Multivariate Integrals	2
4.1	Double Integrals	2
	Riemann Sums to Approximate a Double Integral	2
4.2	Iterative Integration	3
	Partial Integration	3
4.3	Double Integrals as Volume	4
	Double Integrals over Rectangular Regions	4
	Double Integrals over General Regions	4
	Double Integrals in Polar Coordinates	4
4.4	Applications of Double Integrals	4
	Surface Area	4
4.5	Triple Integrals	4
4.6	Interpretations of Triple Integrals	4
	Triple Integrals in Cylindrical Coordinates	4
	Triple Integrals in Spherical Coordinates	4
4.7	The Jacobian	4

4 Multivariate Integrals

4.1 Double Integrals

- **Recall:** If we have a function $f(x)$, we can find the area “under the curve” over the interval $[a, b]$ by computing the integral $\int_a^b f(x) dx$.
- A **double integral** can be used to find the *volume* “under the surface” defined by $z = f(x, y)$.

Riemann Sums to Approximate a Double Integral

Recall: The *Riemann sum* used to approximate the area under $f(x)$ from $x = a$ to $x = b$ is:

$$A \approx \sum_{k=0}^n f(x_k) \Delta x$$

where $\Delta x = \frac{b-a}{n}$, the width of each subinterval, and x_k is a “sample point” in the k^{th} subinterval.

Riemann Sum for Two Variables

1. Define a region and partition

Let R be a rectangular region in the xy -plane defined by $R = [a, b] \times [c, d]$. We can partition R into m subintervals along the x -axis and n subintervals along the y -axis, creating a grid of rectangles.

If we use equal subintervals, we can call $\Delta x = \frac{b-a}{m}$ and $\Delta y = \frac{d-c}{n}$.

2. Define sample points

For each subrectangle, we can choose a sample point (x_i, y_j) within that subrectangle. Then the approximate height to the surface is given by $z_{ij} = f(x_i, y_j)$.

For example, we could choose the sample point to be the lower-left corner of each subrectangle, meaning $x_i = a + i\Delta x$ and $y_j = c + j\Delta y$.

3. Define volume of each box

The area of each subrectangle is given by $\Delta A = \Delta x \Delta y$.

The volume of the box approximating the volume under the surface over each subrectangle is thus $f(x_i, y_j) \Delta A = f(x_i, y_j) \Delta x \Delta y$.

4. Definition of Riemann sum

The **Riemann sum** approximating the volume under the surface over the region R is given by summing the volumes of all the boxes:

$$V \approx \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta A = \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta x \Delta y.$$

The definite integral of $f(x)$ from $x = a$ to $x = b$ (the actual area) is when $n \rightarrow \infty$:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(x_k) \Delta x.$$

Limit Definition of a Double Integral

The double integral of $f(x, y)$ over the rectangular region $R = [a, b] \times [c, d]$ is defined as:

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta A$$

where $\Delta A = \Delta x \Delta y$ is the area of each subrectangle, and (x_i, y_j) is a sample point in each subrectangle.

It is simply when we take infinitely many partitions using the Riemann sum.

Example: 3D Riemann Sum

Use a Riemann Sum to approximate the volume under the surface $z = 9xe^{-xy}$ over the rectangular region $R = [0, 2] \times [0, 1]$ using 2 subintervals in each direction and choosing the sample points to be the upper-right corners of each subrectangle.

The upper-right corners of each subrectangle are given by:

4.2 Iterative Integration

Fubini's Theorem

If $f(x, y)$ is continuous on the rectangle $R = [a, b] \times [c, d]$, then the double integral of f over R can be computed as an iterated integral in either order:

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

Similarly, if we can write $f(x, y)$ as a product of functions of x and y , then we can separate the double integral:

$$\int_a^b \int_c^d g(x)h(y) dy dx = \int_a^b g(x) dx \int_c^d h(y) dy.$$

Partial Integration

- The indefinite integral of a function with multiple variables can be computed by integrating with respect to one variable at a time, treating the other variables as constants. This is known as **partial integration**.
- For example, to compute the partial integral of $f(x, y)$ with respect to x :

$$\int f(x, y) dx = F(x, y) + C(y)$$

where $C(y)$ is an arbitrary function of y .

- Similarly, to compute the partial integral of $f(x, y)$ with respect to y :

$$\int f(x, y) dy = G(x, y) + D(x)$$

where $D(x)$ is an arbitrary function of x .

4.3 Double Integrals as Volume

Double Integrals over Rectangular Regions

- The volume under the surface $z = f(x, y)$ over a rectangular region $R = [a, b] \times [c, d]$ is given by the double integral:

$$V = \int_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx$$

Double Integrals over General Regions

Double Integrals in Polar Coordinates

4.4 Applications of Double Integrals

Surface Area

4.5 Triple Integrals

4.6 Interpretations of Triple Integrals

Triple Integrals in Cylindrical Coordinates

Triple Integrals in Spherical Coordinates

4.7 The Jacobian